ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Hierarchical Clustering

Session 23 and 24

BSDSBA 2028

15 April 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Assessed – Midterm Exam, E3, E4

Due this Week – MP1 Peer Eval (Fri April 17), ICA10 (Wed April 15), E7 (Fri April 17)

Upcoming – R11 (Wed April 15), E4S2 (Fri April 24), E8 (Thu April 30), A2 – Data Mining 1 (Thu April 30), A3 – Data Mining 2 (Fri May 8), MP2 (TBA)

For Checking (Next) – A1, MP1, E3S2

Final Project – Topic Proposal (Fri April 17)

Session 23 and 24 – Hierarchical Clustering

Gameplan

Session 23

Introduction to Hierarchical Clustering

Agglomerative & Divisive Methods

Session 24

Applied Example of Hierarchical Clustering

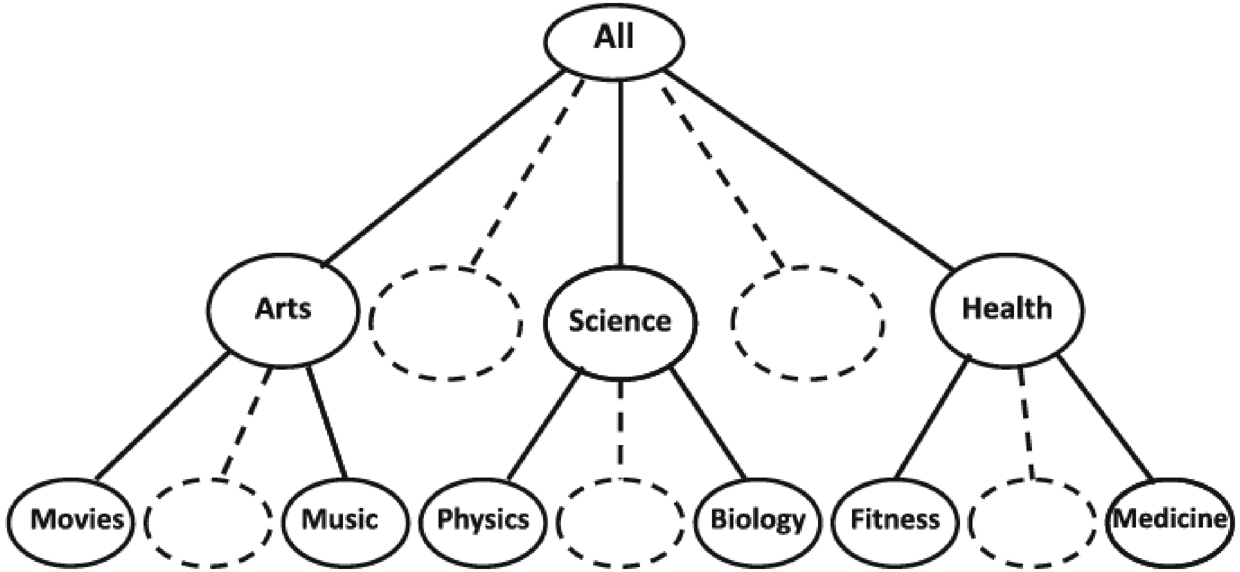
In-Class Activity 11 – Hierarchical Clustering

Introduction to Hierarchical Clustering

Introduction to Hierarchical Clustering

What is Hierarchical Clustering?

A hierarchical clustering method groups data objects into a hierarchy or a "tree" of clusters.



Open Directory Project - https://en.wikipedia.org/wiki/DMOZ

Introduction to Hierarchical Clustering

What are applications of Hierarchical Clustering?

Employee Organization – Company employees may be organized according to major groups (such as executives, managers, and staff).

Handwritten Digits – Handwritten digits may be first partitioned into general groups then into a much finer group depending on the writing style of each number.

Genomics/DNA Analysis – Genes or DNA sequences clustered based on similarity.

Recommender Systems – Items can be categorized into genres, subgenres, then more niche categories.

Introduction to Hierarchical Clustering

What are the different types of Hierarchical Clustering?

Bottom-up (Agglomerative) – Individual data points are agglomerated into higher-level clustered. Clusters are merged according to some similarity measure.

Top-down (Divisive) – Starts by placing all data objects into one cluster (root cluster), then divides the root cluster into several small subclusters.

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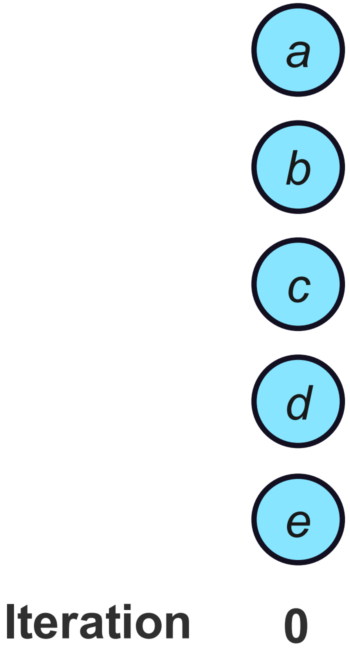

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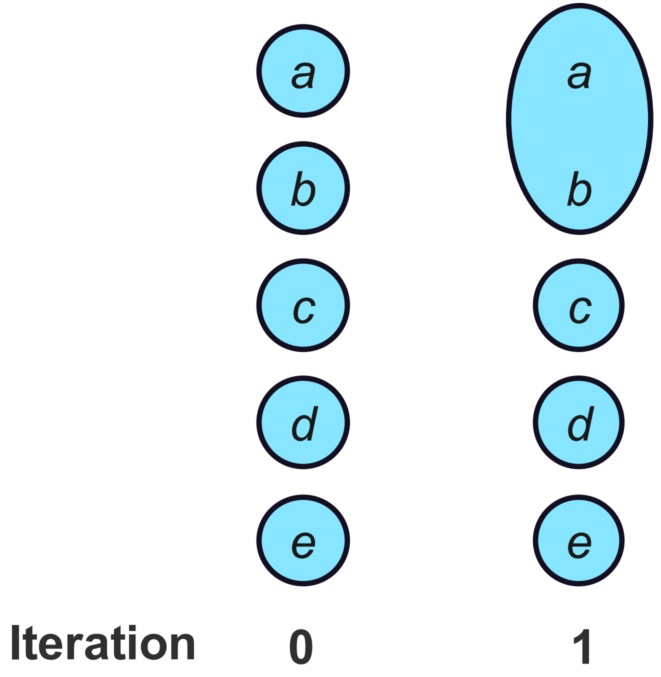

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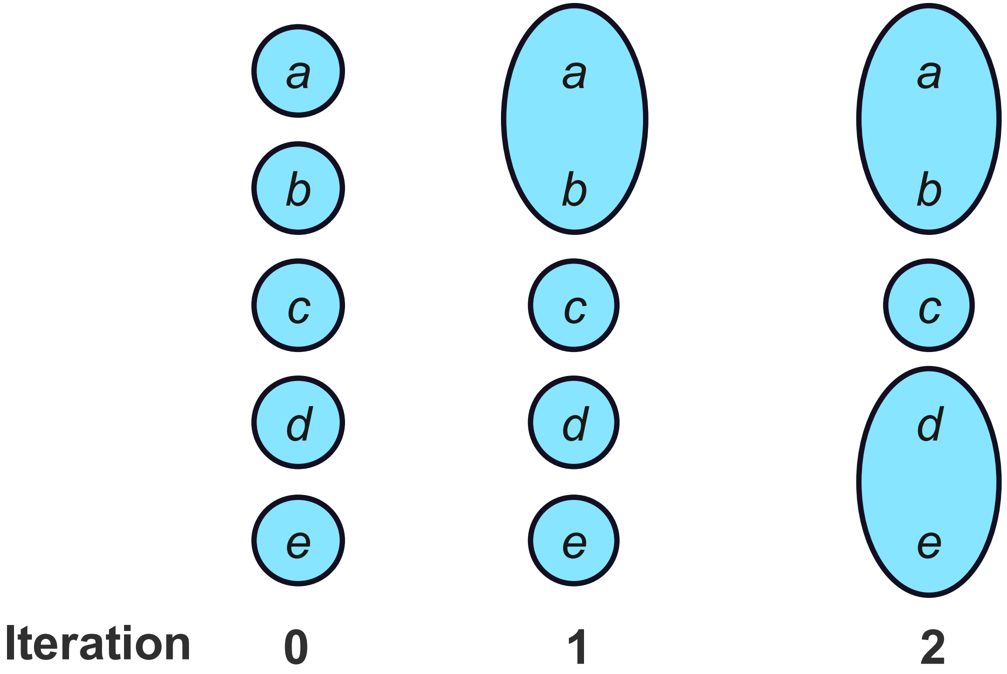

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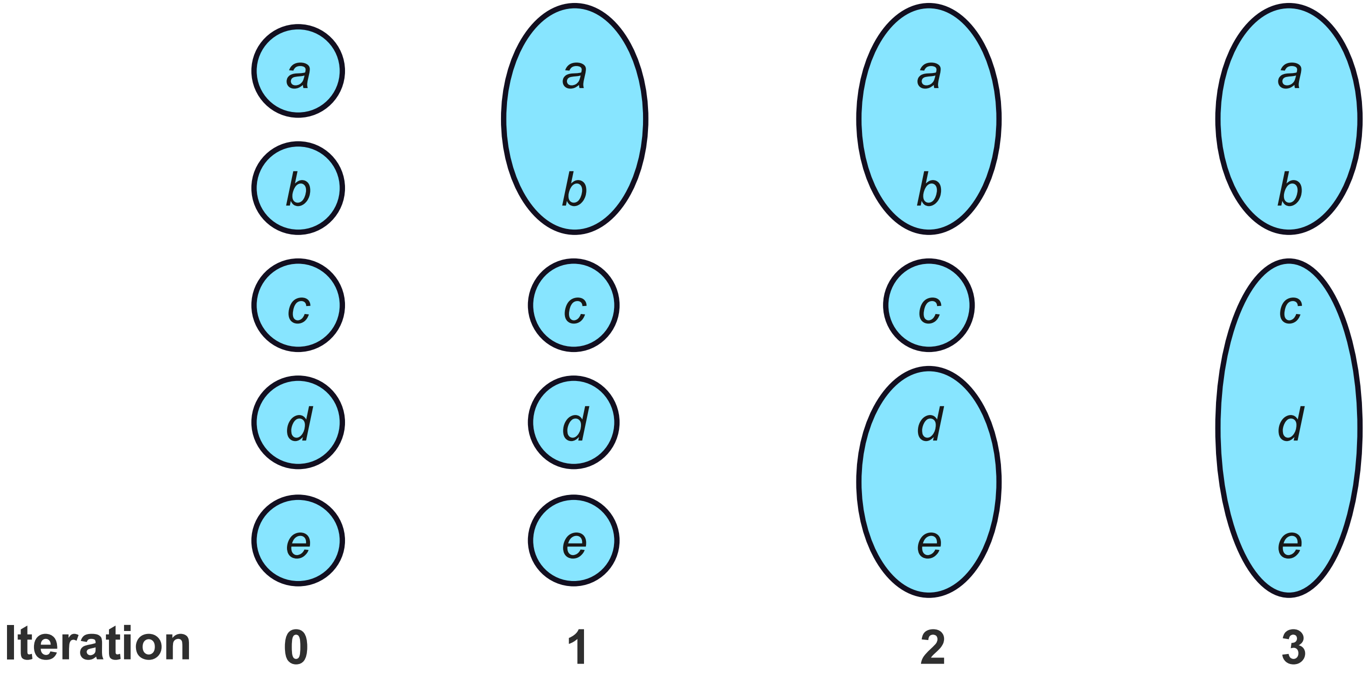

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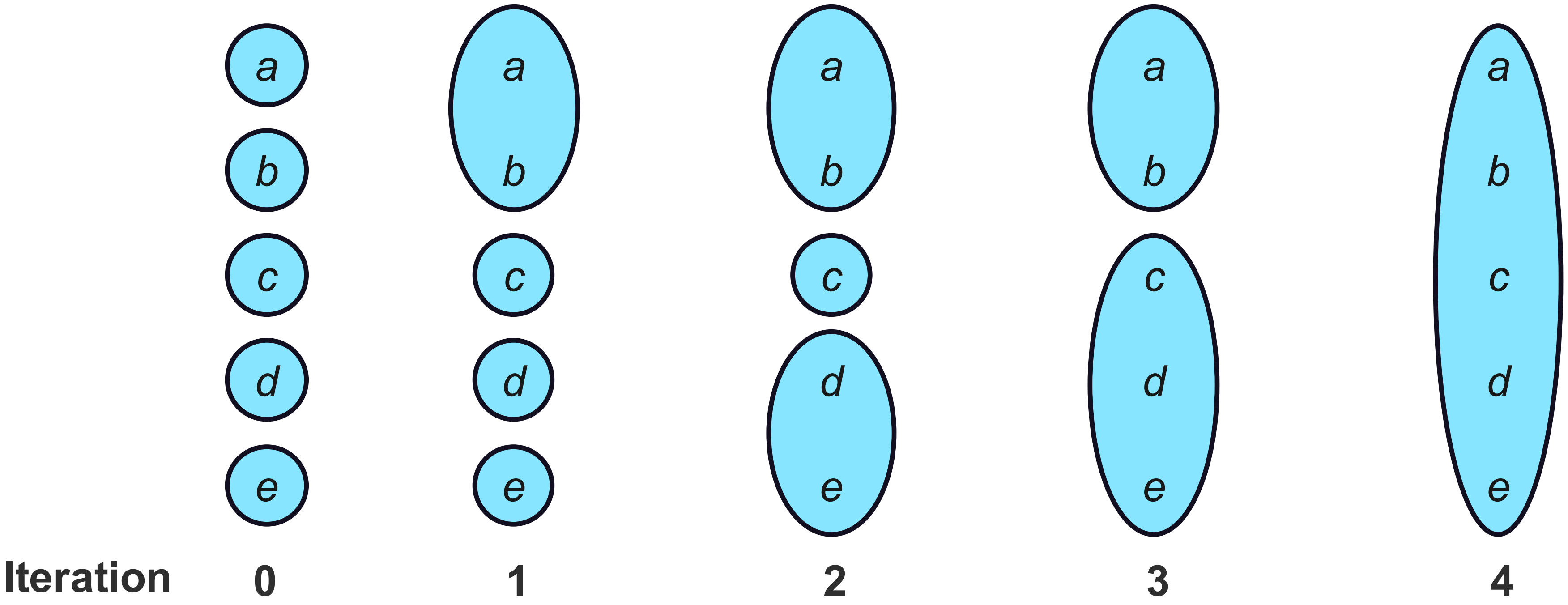

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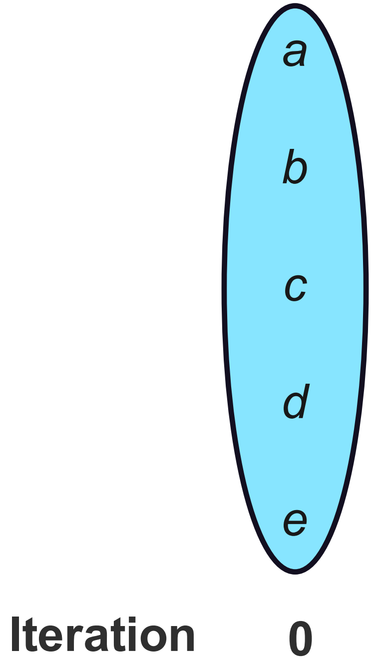

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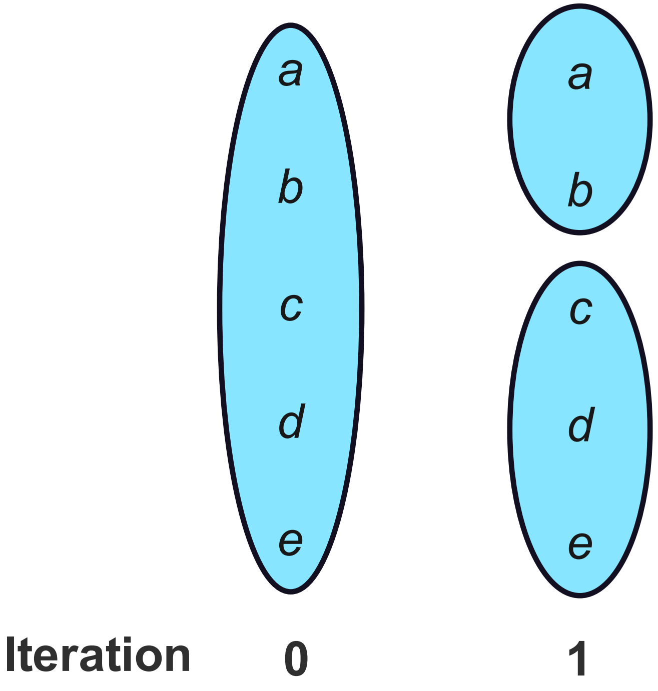

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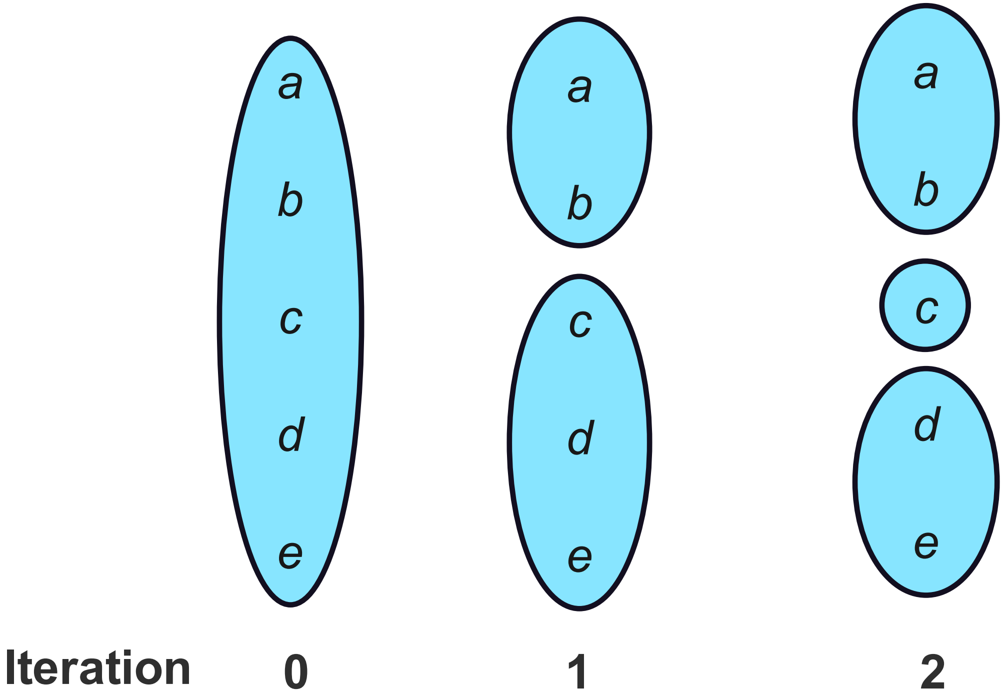

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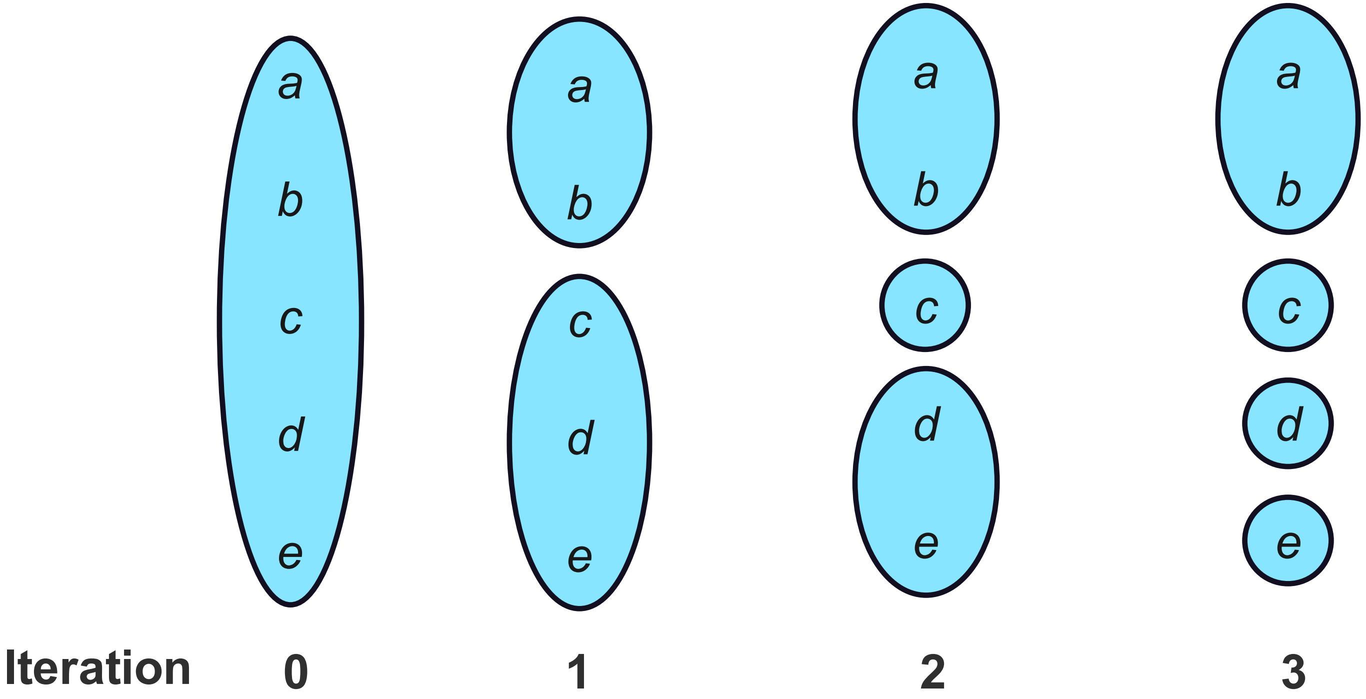

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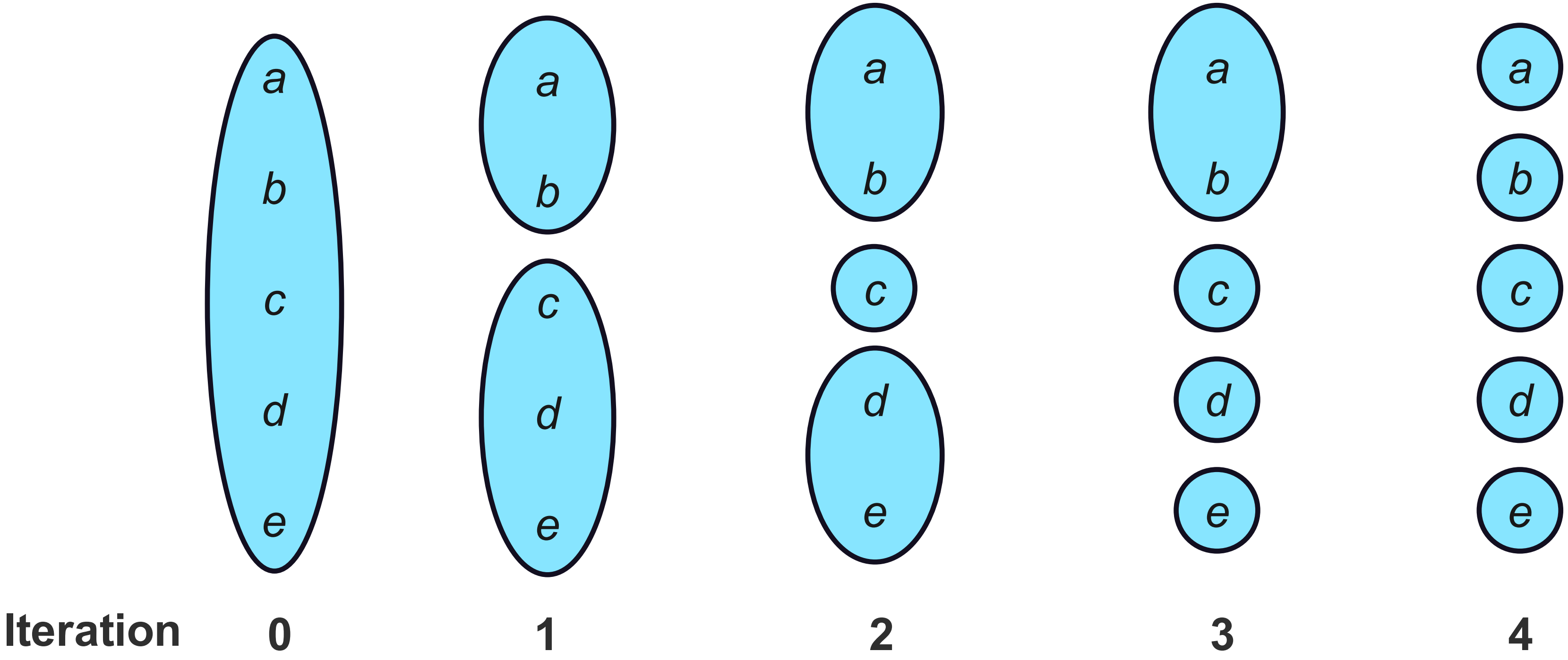


Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering

Similarity Measures in Agglomerative Clustering

Let $\|p - p'\|$ be the distance between two objects or points, p and p'; m_i be the mean of the cluster C_i; and n_i is the number of objects in C_i

Minimum distance (single-linkage): $dist_{min}(C_i, C_j) = \min_{p \in C_i, p' \in C_j} \{\|p - p'\|\}$

Max distance (complete-linkage): $dist_{max}(C_i, C_j) = \max_{p \in C_i, p' \in C_j} \{\|p - p'\|\}$

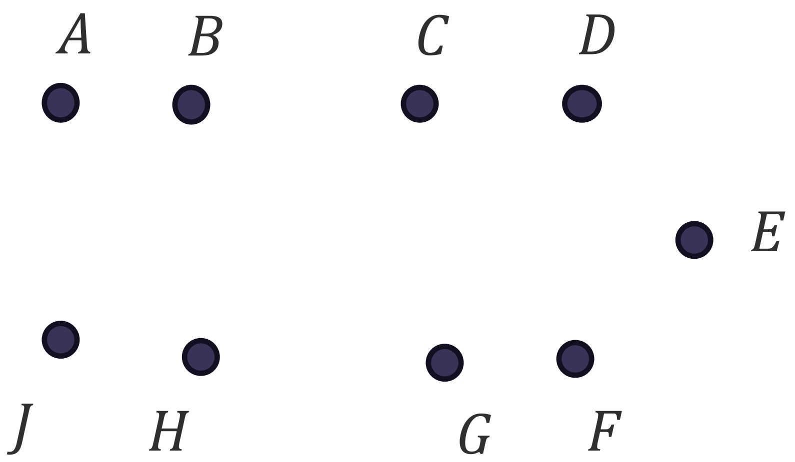
Average distance (group average-linkage): $dist_{avg}(C_i, C_j) = \frac{1}{n_i n_j} \sum_{p \in C_i, p' \in C_j} \|p - p'\|$

Mean distance (centroid clustering): $dist_{mean}(C_i, C_j) = \|m_i - m_j\|$

Agglomerative Hierarchical Clustering

Single vs. Complete Linkages

Single-linkage Clustering

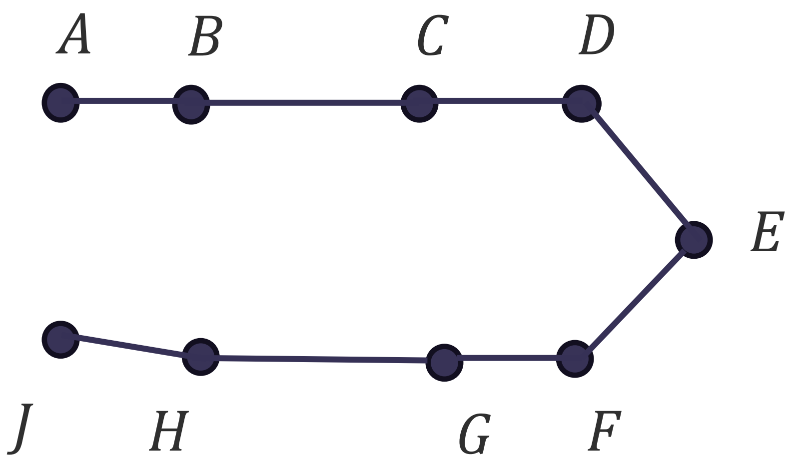


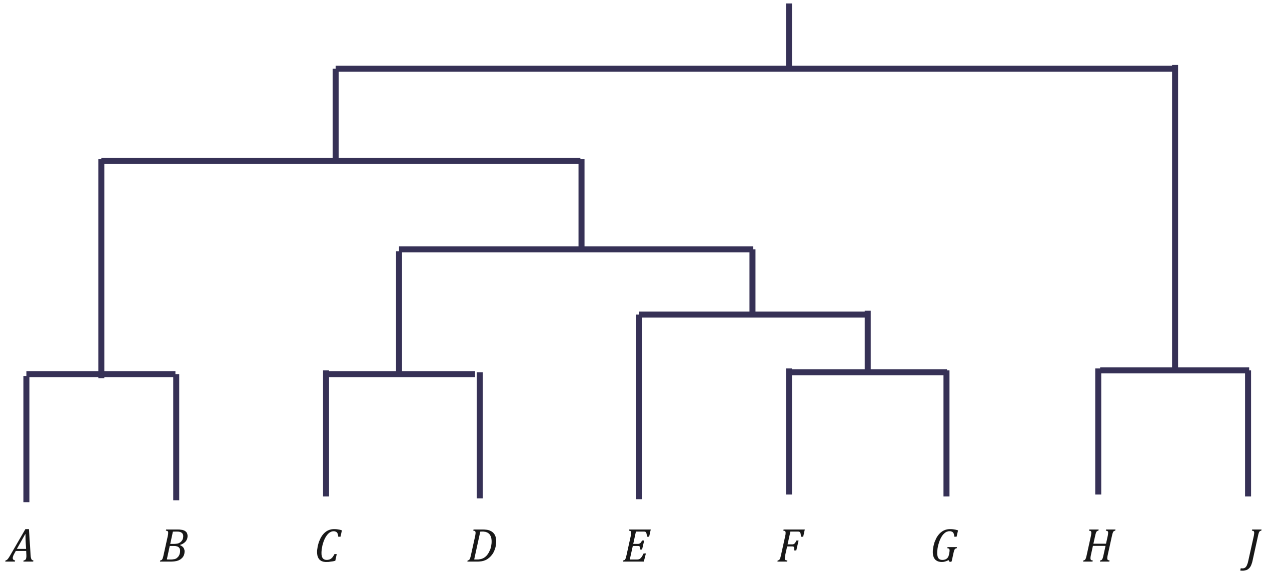
A B C D E F G H J

Agglomerative Hierarchical Clustering

Single vs. Complete Linkages

Single-linkage Clustering

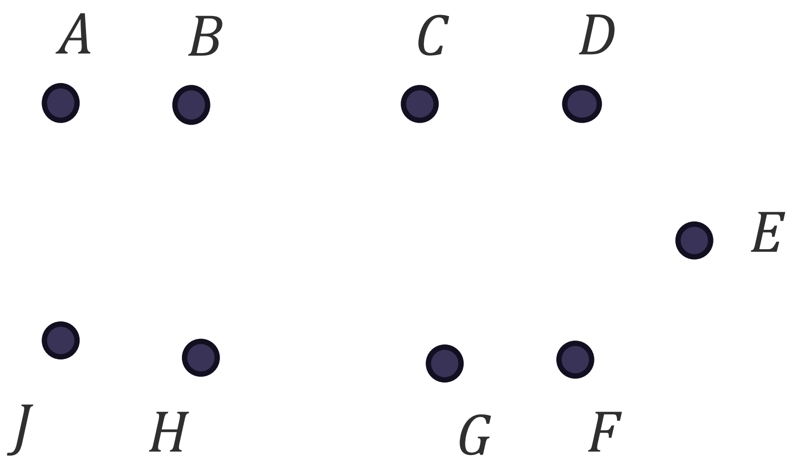




Agglomerative Hierarchical Clustering

Single vs. Complete Linkages

Complete-linkage Clustering

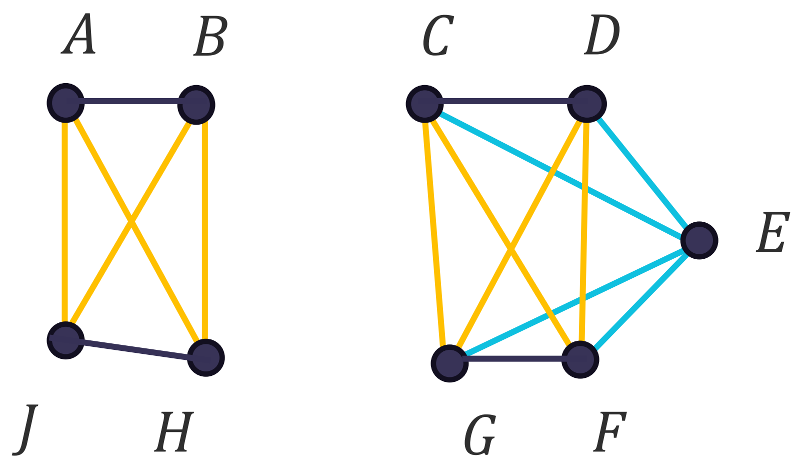


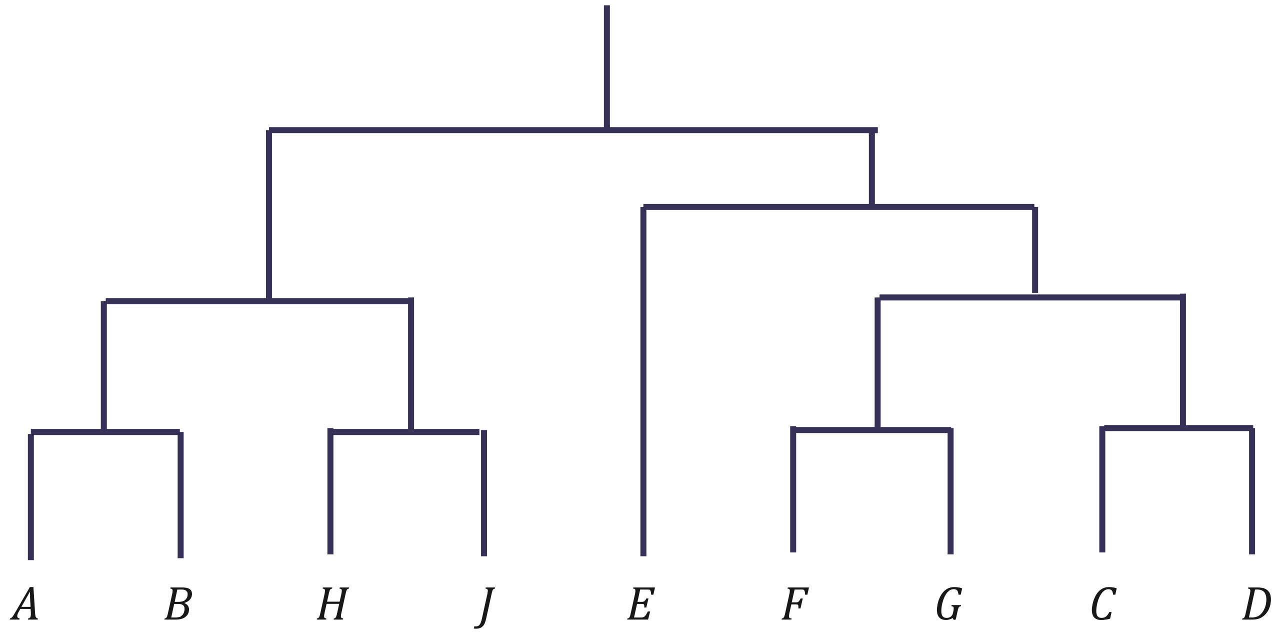
$A \quad B \quad H \quad J \quad E \quad F \quad G \quad C \quad D$

Agglomerative Hierarchical Clustering

Single vs. Complete Linkages

Complete-linkage Clustering





Agglomerative Hierarchical Clustering

Ward's Method

We can also use the increase of the *inertia* or the *within-cluster variation* as condition for merging two clusters. This is known as ***Ward's Criterion*** or ***Ward's Method.***

Suppose two disjoint clusters C_i and C_j are merged, and $m_{(ij)}$ is the mean of the new cluster.

$$W(C_i, C_j) = \sum_{x \in C_i \cup C_j} \left\| x - m_{(ij)} \right\|^2 - \sum_{x \in C_i} \|x - m_i\|^2 - \sum_{x \in C_j} \left\| x - m_j \right\|^2$$

$$= \frac{n_i n_j}{n_i + n_j} \left\| m_i - m_j \right\|^2$$

Divisive
Hierarchical Clustering

Divisive Hierarchical Clustering

Algorithm for Divisive Hierarchical Clustering

Given a data set $\mathcal{D}$ and a clustering algorithm $\mathcal{A}$

start

Initialize tree $\mathcal{T}$ to root containing $\mathcal{D}$;

repeat

Select a leaf node L in $\mathcal{T}$ based on pre-defined criterion;

Use algorithm $\mathcal{A}$ to split L into $L_1, \dots, L_k$

Add $L_1, \dots, L_k$ as children of L in $\mathcal{T}$;

until termination condition

end

Divisive Hierarchical Clustering

Bisecting k-Means

Given a data set $\mathcal{D}$ and then using the k-Means clustering algorithm as $\mathcal{A}$

start

Initialize tree $\mathcal{T}$ to root containing $\mathcal{D}$;

repeat

Select a leaf node L in $\mathcal{T}$ based on pre-defined criterion;

Use algorithm $\mathcal{A}$ to split L into L_1 and L_2

Add L_1 and L_2 as children of L in $\mathcal{T}$;

until termination condition

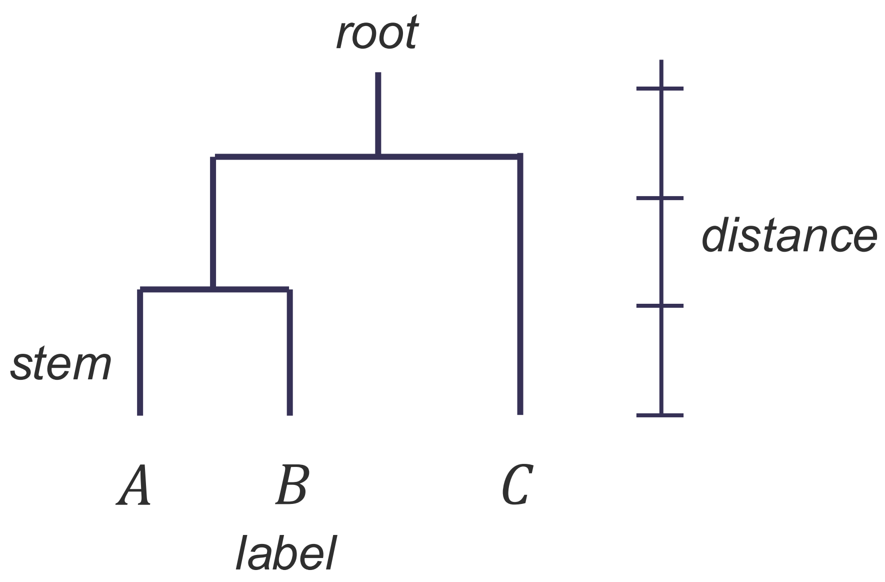
end

Applying Hierarchical Clustering

Applying Hierarchical Clustering

Dendrogram or Tree Diagram

The dendrogram is a mathematical and pictorial representation of the complete clustering procedure.



Node – represent clusters

Length of Stems (Height) – represent the distances at which clusters are joined

Topology – arrangement of nodes and stems

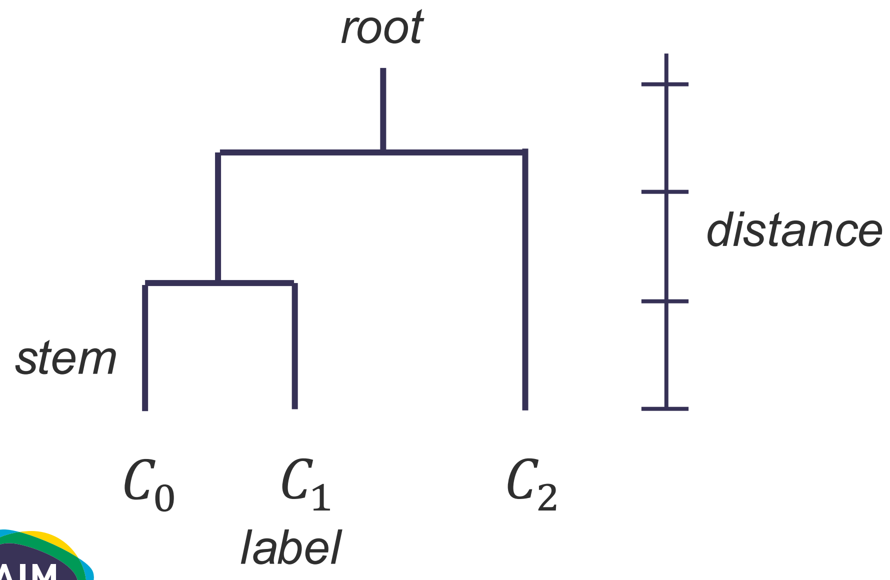
Exemplars or **Centrotypes** – objects having the maximum within-cluster average similarity

Medoids – *centrotype* with minimum absolute distance to the other members of the cluster

Applying Hierarchical Clustering

Dendrogram or Tree Diagram

An $(n-1) \times 4$ **linkage matrix Z** describes how hierarchical clusters are created. Z shows which clusters were merged, the distance between them, and the size of the new cluster.

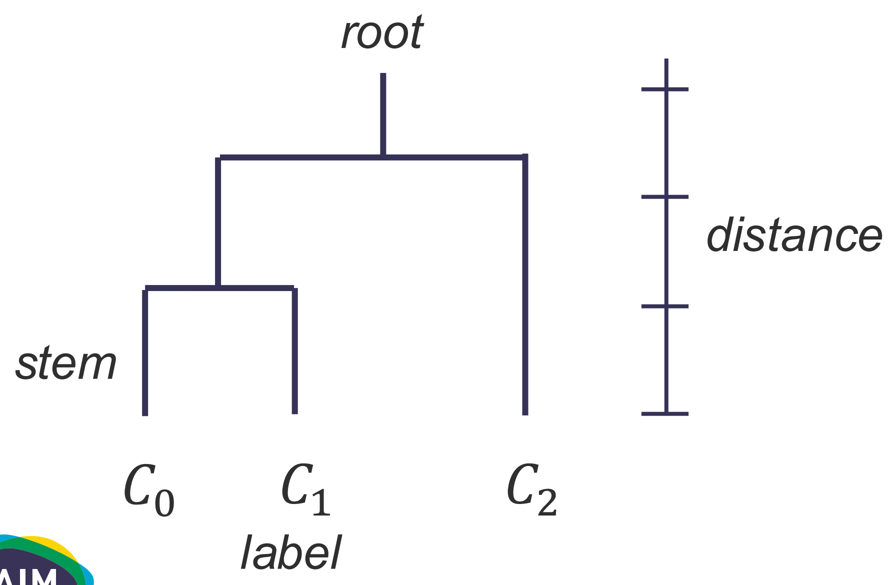


$$Z = \begin{bmatrix} C_0 & C_1 & dist(C_0, C_1) & 2 \\ C_2 & C_3 & dist(C_2, C_3) & 3 \end{bmatrix}$$

Applying Hierarchical Clustering

Comparing Dendrograms

A commonly used method for comparing dendrograms is the computation of the ***cophenetic correlation***. Or the correlation between the original proximity matrix D and the ***cophenetic matrix*** $\boldsymbol{H}$.



$\boldsymbol{H}$ contains the heights $\boldsymbol{h_{ij}}$ where two objects become members of the same cluster

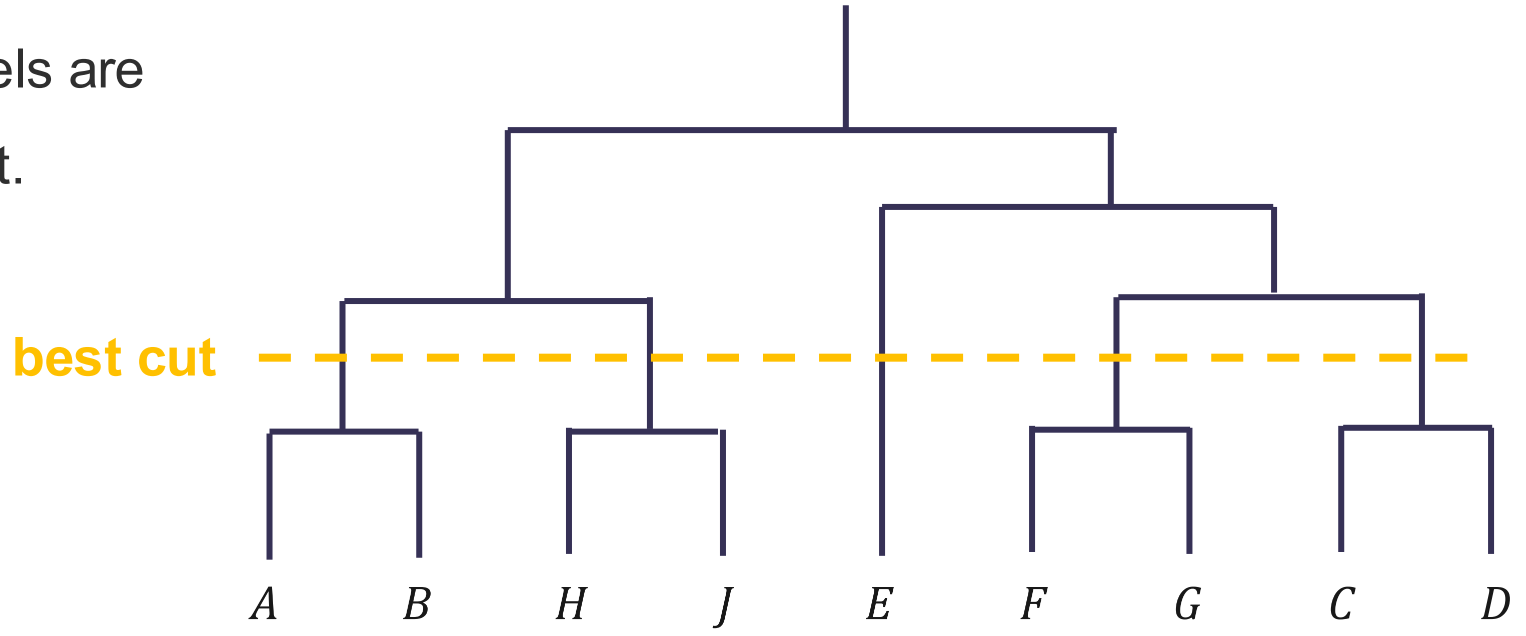
$$H = \begin{bmatrix} 0 & & \\ h_{01} & 0 & \\ h_{02} & h_{12} & 0 \end{bmatrix}$$

Applying Hierarchical Clustering

Choosing the Number of Cluster

In hierarchical clustering, partitions are achieved by selecting a particular height to cut a dendrogram to (termed as the *best cut**).

*large changes in fusion levels are taken to indicate the best cut.



Applying Hierarchical Clustering

Choosing the Number of Cluster

Other 'number of cluster' procedures have also been suggested, one of which is the ***upper tail rule*** which proposed to select the number of clusters as the first stage in the dendrogram that satisfies:

$$\alpha_{j+1} > \overline{\alpha} + k s_{\alpha}$$

where $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$ are the fusion levels corresponding to stages with $n, n-1, \ldots, 1$ clusters. $\overline{\alpha}$ and s_{α} are the mean and standard deviation of the j previous levels, and k is a constant.

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Applying Hierarchical Clustering

Advantages

- Hierarchical clustering methods provides different levels of clustering granularity which has the potential to give application specific-insights.
- Some hierarchical clustering methods can discover clusters of arbitrary shape.
- Hierarchical clustering methods (esp. divisive methods) provide a lot of control to the user. They can be combined with other clustering techniques.

Applying Hierarchical Clustering

Disadvantages

- Unclear as to how to determine the *best cut* for partitioning data into clusters.
- Susceptible to noise (especially single-linkages), error propagates quickly if incorrect merge or split points have been made.
- Methods do not scale well with dataset size. If similarity matrix M cannot be incrementally maintained, most hierarchical methods will have a time complexity that increases to $O(n^3 d)$.